Code No: 133BD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, April/May - 2023
MATHEMATICS - IV
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, MIE, CEE, MSNT)
Time: 3 Hours **Max. Marks: 75**

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Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Define analytic function. [2]
b) Find the limit of the function $\lim_{z\to 2i}(3x + iy^2)$. [3]
c) Determine the poles of the integral $\int_c \frac{4-3z}{z(2-z)}dz$. [2]
d) Explain about isolated singularity and removable singularity. [3]
e) Write the cross ratio of four points z_1, z_2, z_3, z_4. [2]
f) Find u and v values of the bilinear transformation $w = coshz$. [3]
g) Define Fourier integral theorem. [2]
h) Define the periodic function. [3]
i) Write down the one-dimensional wave equation. [2]
j) Explain the method of separation of variables. [3]

PART – B

(50 Marks)

2.a) Check the continuity of $f(z) = \begin{cases} \frac{z}{|z|} & if\ z\neq 0 \\ 0 & if\ z=0 \end{cases}$.
b) Discuss the analyticity of the function $f(z) = z\bar{z}$. [5+5]

OR

3.a) Show that $u = \frac{1}{2}\log(x^2 + y^2)$ is harmonic and find its harmonic conjugate function.
b) Using definition of derivative, find $f'(z)$, if it exists for the following function $f(z) = \frac{1+z}{(1-z)}, z \neq 1$. [5+5]

4.a) Find the residue of $f(z) = \frac{z^3}{z^2-1}$ at $z = \infty$.
b) Evaluate $\oint_c \frac{e^{2z}}{(z+i)^4}dz$ where C is the circle $|z| = 3$. [5+5]

OR

5.a) Evaluate the complex integral $\oint_c tanz\ dz$ where C is the circle $|z| = 2$.
b) Find the Taylor's series expansion for the function $f(z) = \left(\frac{z}{z+1}\right)^2$ $z_0 = 1$ about the point z_0. [5+5]

6.a) Using contour integration method prove the integral $\int_0^{2\pi} \frac{x^{a-1}}{1+x}\, dx = \frac{\pi}{sin\pi a}$.

b) Find the bilinear transformation such that the point z=i is the only fixed point and the point '1' is mapped as ∞. [5+5]

OR

7.a) Discuss fully the transformation $w = c\ coshz$, where c is a real number. What physical problem can we study with the help of this transformation?

b) Evaluate $\int_0^{\infty} \frac{cos3\theta}{5+4cos\theta}\, d\theta$. [5+5]

8.a) Evaluate the Fourier series expansion of $f(x) = e^{-ax}$ in $(0, 2\pi)$.

b) Find Fourier transform of $f(x) = xe^{-x}, in\ 0 \le x \le \infty$. [5+5]

OR

9.a) Find Fourier sine and cosine transform of $f(x) = 2x$ in $0 < x < 4$.

b) Find Fourier series expansion of $f(x) = \frac{(\pi - x)}{2}\ in\ 0 < x < 2$. [5+5]

10.a) Solve the equation by method of separation of variables $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$ and $u(0, y) = e^{-5y}$.

b) Explain the possible solutions of one-dimensional heat equation. [5+5]

OR

11. A tightly stretched string with fixed end points x=0 and x=l is initially in a position given $y(x, 0) = y_o sin^3(\frac{\pi x}{l})$. If it is released from rest position, find the displacement y at any time and any distance from the end x=0. [10]

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